

Education

University of Pennsylvania	Aug 2025 - Expected May 2027
MAS-CS, Computer Science	
Relevant Coursework: Artificial Intelligence, Machine Learning, Distributed Systems, Algorithms, Data Structures, Computer Systems, Database Systems	

Technical Skills

- **Applied AI:** Agentic AI Systems, RAG, LLM Evaluation, LLM Observability, Embeddings, Hybrid Search, Reranking, MCP, Prompt Engineering
- **Languages & Backend:** Python, JavaScript/TypeScript, Java, SQL, Node.js, Flask, Express, REST APIs
- **Data & Observability:** Prometheus, ELK, OpenTelemetry, pgvector, Oracle DB, PostgreSQL, MySQL, Spark, Airflow
- **Cloud & Infrastructure:** AWS, Docker, Kubernetes, Linux, GitLab CI, Helm, Distributed Systems, TCP/IP

Work Experience

TSMC	Jun 2026 - Jul 2026
AI Engineer Intern, Infrastructure and Platform Development	
• Built an end-to-end AI agent root cause analysis system using OpenCode and DeepSeek-V4 across 10 fabs, reducing troubleshooting time from 2–3 hours to under 5 minutes • Developed 10 custom MCP tools on a unified Skill framework, using persistent TCP connections to ingest Prometheus, ELK, and Oracle DB telemetry and reducing end-to-end retrieval latency by 90% • Designed a hybrid decision pipeline combining 7-day rolling-statistics baselines with LLM reasoning to adjust alert thresholds dynamically, reducing false alarms by 70% • Implemented a semantic-search RAG knowledge base that retrieved relevant troubleshooting guides to ground LLM diagnoses	
Tesla	Sep 2024 - Mar 2025
Sales Analyst	
• Built an internal SPA with REST services and developed SQL/Tableau KPI workflows that guided sales-process redesign, contributing to a 7.6% regional sales increase	
Shopee, Sea Group	Oct 2023 - Apr 2024
Software Engineer Intern	
• Built an asynchronous task scheduler with Celery, RabbitMQ, and PostgreSQL, reducing manual warehouse scheduling workload by over 90% • Migrated ETL workflows to an Airflow-orchestrated Spark pipeline on Kubernetes, improving scalability and reducing compute costs by 75% • Automated Kubernetes deployments with GitLab CI, Docker, and Helm, reducing infrastructure costs by 50%	
Microsoft	Jun 2022 - Jun 2023
Data Engineer Intern	
• Shipped Flask microservices integrating budgeting and CRM systems, reducing manual reconciliation by 83% and enabling near-real-time visibility • Automated reporting through Power BI and Azure REST APIs and built Python data pipelines that supported outreach converting 20+ customers	

Selected Projects

JobTrail – Agentic Job-Search Intelligence Platform	JavaScript, Node.js, Express, SQLite, OpenAI API, Gmail API, Chrome Extension	Jul 2026
• Built an AI-driven, local-first job-search platform that unifies Gmail application updates, Chrome-captured job descriptions, category-aware resume matching, and funnel analytics in one workflow • Engineered structured LLM pipelines with schema-validated email classification, retry handling, token logging, and cached JD skill extraction, then ranked category-specific resumes using a 60/25/15 skills-and-experience rubric • Limited LLM-bound email content to 1,500 characters and validated duplicate handling, state transitions, and failure recovery with 46 offline tests		
YouTube Music DJ – LLM-Powered Recommendation Agent	JavaScript, OpenAI API, YouTube Data API, Chrome Extension	May 2026
• Built a bilingual AI DJ that converts natural-language moods into controllable, personalized 10-track queues without manual curation • Engineered a GPT-4o-mini structured-output pipeline that converts listening intent into editable search specifications and applies constraint-aware retrieval, ranking, caching, and fallback search • Added feedback-driven re-curation and persistent queue state through a Manifest V3 service worker		
SafeTrip – Real-Time Travel Risk Analytics Platform	Python, React, Node.js, MySQL, AWS, Google Maps API	Dec 2025
• Built a travel-safety platform processing 500K+ NYC Open Data and Airbnb records to generate personalized map recommendations • Designed a BCNF MySQL schema and improved query performance through indexing, query-plan analysis, and API caching		